UNIVERSAL INJECTIVITY OF A FLAT BASE CHANGE HITTING ALL RELEVANT ASSOCIATED PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The stalkwise criterion for universal injectivity and the local version of the statement. Algebra, label algebra-lemma-universally-injective-check-stalks (proof line 1927); Same chapter flat.tex, label lemma-base-change-universally-flat-local (proof line 1931). The first reduces universal injectivity to the localisations at the primes of the target, and the second supplies the local statement for the localised ring maps and the localised module.

Source reference. Original lemma 05CL: sources/stacks-chow/tags.txt, source lines 6924–6924. Navigation pointer for \ref{algebra-lemma-universally-injective-check-stalks} used inside the counted span (line 1927); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DG: sources/stacks-flat/pinned-original-source.tex, source lines 1848–1848. Navigation pointer for \ref{lemma-base-change-universally-flat-local} used inside the counted span (line 1931); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let N be an S-module. Let $S \to S'$ be a ring map. Assume*

(1) *$R \to S$ is of finite presentation and N is of finite presentation over S,*
(2) *N is flat over R,*
(3) *$S \to S'$ is flat, and*
(4) *the image of $\mathrm{Spec}(S') \to \mathrm{Spec}(S)$ contains all primes $\mathfrak{q}$ such that $\mathfrak{q}$ is an associated prime of $N \otimes_R \kappa(\mathfrak{p})$ where $\mathfrak{p}$ is the inverse image of $\mathfrak{q}$ in R.*

Then $N \to N \otimes_S S'$ is R-universally injective.

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Proof. By Algebra, Lemma 05CL it suffices to show that $N_{\mathfrak{q}} \to (N \otimes_R S')_{\mathfrak{q}}$ is a $R_{\mathfrak{p}}$-universally injective for any prime $\mathfrak{q}$ of S lying over $\mathfrak{p}$ in R. Thus we may apply Lemma 05DG to the ring maps $R_{\mathfrak{p}} \to S_{\mathfrak{q}} \to S'_{\mathfrak{q}}$ and the module $N_{\mathfrak{q}}$. $\square$